

Decoder

Working mechanism for <EOS>

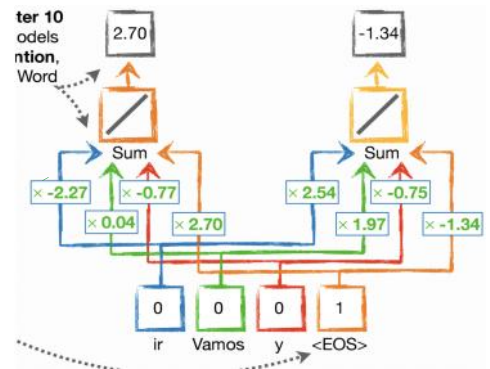
① Embedding values

$$E_{\langle \text{EOS} \rangle} = [2.7 \quad -1.34]$$

You can find it by taking one-hot vector

over the vocabulary = $\begin{bmatrix} \text{ir} \\ \text{Vamos} \\ y \\ \langle \text{EOS} \rangle \end{bmatrix}$

[at first there is no predicted word at the decoder side. So we start with a dummy token <EOS>]



② Positional Encoding

$$d_{\text{model}} = 2$$

For <EOS>

$$E_{\langle \text{EOS} \rangle} = [2.7 \quad -1.34]$$

$$\text{pos} = 0 \quad [\text{first word in the sentence, pos} = 0]$$

$$i = 0, 0 \quad [d_{\text{model}} = 2, \text{ for embedding-1 } i = 0 \\ \text{for embedding-2 } i = 0]$$

$$\begin{aligned} PE(\text{pos} = 0, 2 \times i = 0) &= \sin\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{model}}}}}\right) \\ [\text{for } 2.7] &= \sin\left(\frac{0}{10000^{\frac{2 \times 0}{2}}}\right) = 0 \quad [\text{in Radian}] \end{aligned}$$

$$\begin{aligned} PE(\text{pos} = 0, 2 \times i + 1 = 1) &= \cos\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{model}}}}}\right) \\ [\text{for } -1.34] &= \cos\left(\frac{0}{10000^{\frac{2 \times 0}{2}}}\right) = 1 \quad [\text{in Radian}] \end{aligned}$$

Embedding of <EOS> after adding P.E.

$$E_{\langle \text{EOS} \rangle} = [2.7 \quad -1.34] + [0 \quad 1] = [2.7 \quad -0.34]$$

③ Calculate Self attention: [There is only $\langle \text{Eos} \rangle$ now]

The Query, Key and value weights

$$Q = \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix} \quad K = \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix} \quad V = \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix}$$

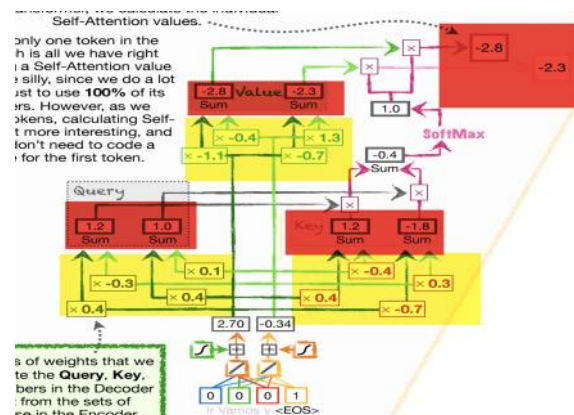
Q, K, V of $\langle \text{Eos} \rangle$

$$E_{\text{Eos}} = [2.7 \quad -0.34]$$

$$Q_{\text{Eos}} = E_{\text{Eos}} \times Q = [2.7 \quad -0.34] \times \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix} = [1.2 \quad 1]$$

$$K_{\text{Eos}} = E_{\text{Eos}} \times K = [2.7 \quad -0.34] \times \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix} = [1.2 \quad -1.8]$$

$$V_{\text{Eos}} = E_{\text{Eos}} \times V = [2.7 \quad -0.34] \times \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix} = [-2.8 \quad -2.3]$$



Attention of $\langle \text{Eos} \rangle$

② We will multiply Query of $\langle \text{Eos} \rangle$ with Key of $\langle \text{Eos} \rangle$ as there is only one token now.

$$Q_{\text{Eos}} \cdot K_{\text{Eos}} = [1.2 \quad 1] \cdot [1.2 \quad -1.8] = (1.2 \times 1.2) + \{1 \times (-1.8)\} = 0.4$$

② Do Softmax to find Similarity to Let's

$$\begin{aligned} & \text{Softmax}(Q_{\langle \text{EOS} \rangle} \cdot K_{\langle \text{EOS} \rangle}) \\ &= \text{softmax}(0.4) \\ &= \left[\frac{e^{0.4}}{e^{0.4}} \right] \\ &= [1] \end{aligned}$$

③ Attention for $\langle \text{EOS} \rangle$

$$V_{\text{EOS}} \times 1 = [-2.8 \quad -2.3]$$

④ Residual connection [update Embedding of $\langle \text{EOS} \rangle$]

$$\begin{aligned} E_{\langle \text{EOS} \rangle} &= E_{\text{EOS}} + \text{Attention of } \langle \text{EOS} \rangle \\ &= [2.7 \quad -0.34] + [-2.8 \quad -2.3] \\ &= [-0.1 \quad -2.6] \end{aligned}$$

④ Calculate Cross Attention

To calculate cross attention we need Query & Value of $\langle \text{EOS} \rangle$ [Query & Value of decoder token] and key & value of $\langle \text{Let's} \rangle$ and key & value of $\langle \text{go} \rangle$ [key & value of encoder tokens]

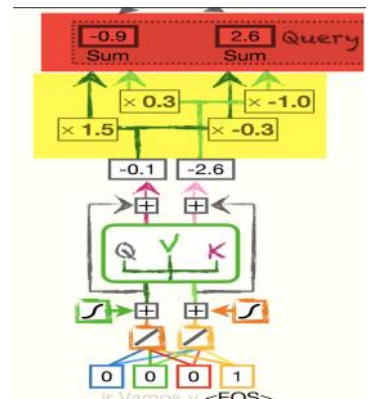
Q, V of $\langle \text{EOS} \rangle$:

Query & Value weights, $Q = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix}$ $V = \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix}$

$E_{\text{EOS}} = [-0.1 \quad -2.6]$ [after self attention]

$$\begin{aligned} Q_{\text{EOS}} &= E_{\text{EOS}} \times Q = [-0.1 \quad -2.6] \times \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix} \\ &= [-0.9 \quad 2.6] \end{aligned}$$

$$\begin{aligned} V_{\text{EOS}} &= E_{\text{EOS}} \times V = [-0.1 \quad -2.6] \times \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix} \\ &= [1.1 \quad 2.1] \end{aligned}$$



$$= \begin{bmatrix} 6 & 3.1 \end{bmatrix}$$

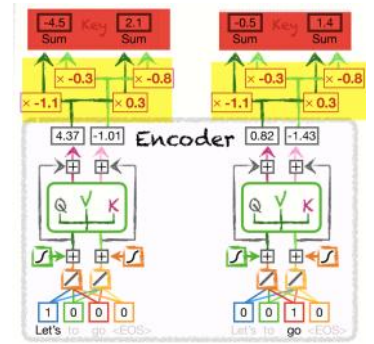
K, V of Lets and go:

key and value weight matrices

$$K = \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix} \quad V = \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix}$$

$$E_{\text{Lets}} = \begin{bmatrix} 4.37 & -1.01 \end{bmatrix} \quad [\text{after self attention at encoder}]$$

$$E_{\text{go}} = \begin{bmatrix} 0.82 & -1.43 \end{bmatrix}$$



$$K_{\text{lets}} = E_{\text{lets}} \times K = \begin{bmatrix} 4.37 & -1.01 \end{bmatrix} \times \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix} = \begin{bmatrix} -4.5 & 2.1 \end{bmatrix}$$

$$K_{\text{go}} = E_{\text{go}} \times K = \begin{bmatrix} 0.82 & -1.43 \end{bmatrix} \times \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix} = \begin{bmatrix} -0.5 & 1.4 \end{bmatrix}$$

$$V_{\text{lets}} = E_{\text{lets}} \times V = \begin{bmatrix} 4.37 & -1.01 \end{bmatrix} \times \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix} = \begin{bmatrix} 6.0 & 3.1 \end{bmatrix}$$

$$V_{\text{go}} = E_{\text{go}} \times V = \begin{bmatrix} 0.82 & -1.43 \end{bmatrix} \times \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix} = \begin{bmatrix} 2.6 & 1.2 \end{bmatrix}$$

Cross attention

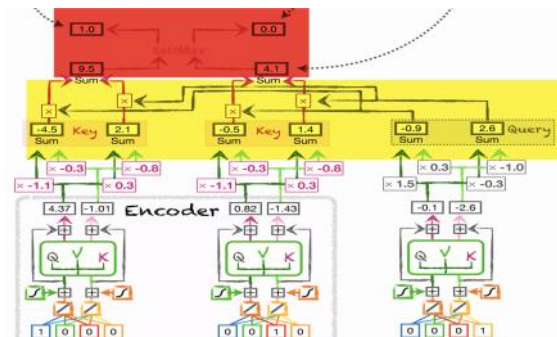
① We will multiply Query of <Eos> [decoder] with the key of Lets and key of go [Encoder]

$$Q_{\text{Eos}} \cdot K_{\text{lets}} = \begin{bmatrix} -0.9 & 2.6 \end{bmatrix} \cdot \begin{bmatrix} -4.5 & 2.1 \end{bmatrix} \\ = (-0.9) \times (-4.5) + (2.6 \times 2.1) = 9.5$$

$$Q_{\text{Eos}} \cdot K_{\text{go}} = \begin{bmatrix} -0.9 & 2.6 \end{bmatrix} \cdot \begin{bmatrix} -0.5 & 1.4 \end{bmatrix} \\ = (-0.9) \times (-0.5) + (2.6 \times 1.4) = 4.1$$

② Do softmax to find similarity to <Eos>

$$\text{softmax}(Q_{\text{Eos}} \cdot K_{\text{lets}}, Q_{\text{Eos}} \cdot K_{\text{go}}) \\ = \text{softmax}(9.5, 4.1)$$

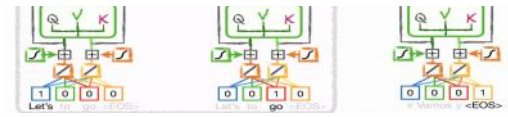


$$\text{softmax} (9.5, 4.1)$$

$$= \text{softmax} (9.5, 4.1)$$

$$= \left[\frac{e^{9.5}}{e^{9.5} + e^{4.1}}, \frac{e^{4.1}}{e^{9.5} + e^{4.1}} \right]$$

$$= [1, 0]$$



③ Attention for

$$(V_{\text{lets}} \times 1) + (V_{\text{go}} \times 0)$$

$$= [6.0 \quad 3.1] \times 1 + [2.6 \quad 1.2] \times 0$$

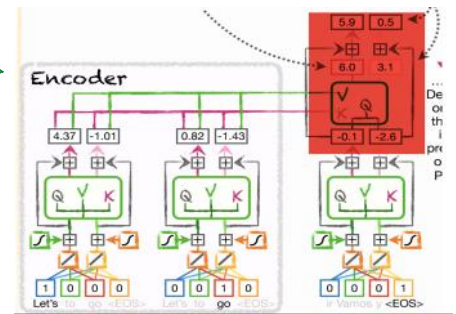
$$= [6.0 \quad 3.1]$$

④ Residual connection [update Embedding of <EOS>]

$$E_{\text{<EOS>}} + \text{Attention of}$$

$$= [-8.1 \quad -2.6] + [6.0 \quad 3.1]$$

$$= [-5.9 \quad 0.5]$$



⑤ Next token prediction

[5.9 and 0.5] gets multiplied with a weight matrix and we get [-5.1 5.7 -3.6 -4.6]

$$\text{softmax} ([-5.1 \quad 5.7 \quad -3.6 \quad -4.6])$$

$$= [0 \quad 1 \quad 0 \quad 0]$$

$$\downarrow$$

$$[\text{ir} \quad \text{Vamos} \quad \text{y} \quad \text{<EOS>}]$$

↳ Vamos gets predicted as next token.

